

3. Merging

3.1 Merging the normal and leukemia datasets

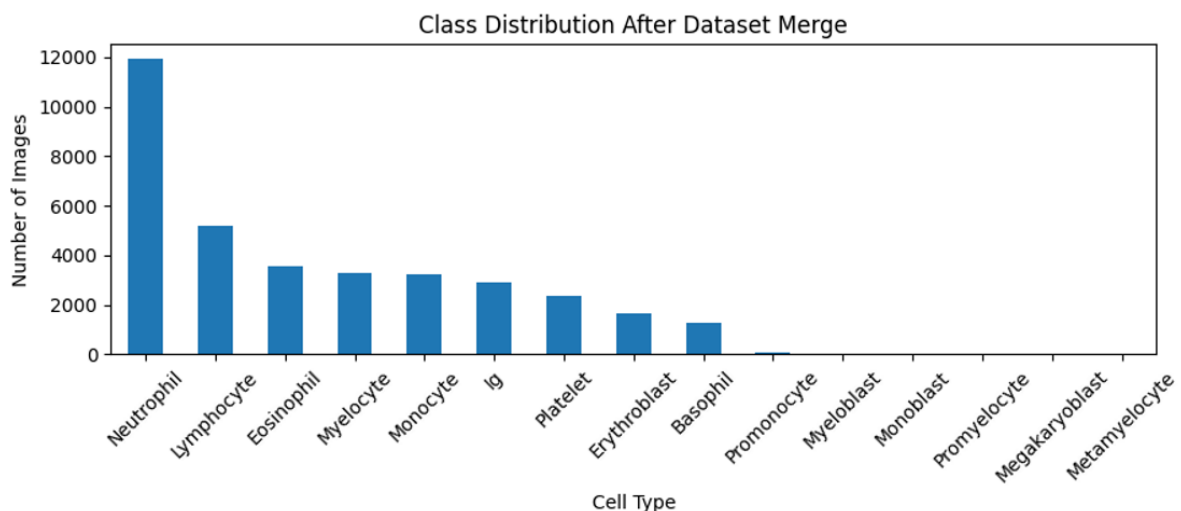
The normal blood cell dataset and the leukemia blood cell dataset were merged to create a more realistic and useful dataset. In real blood samples, normal and abnormal cells appear together, not separately. By merging both datasets, the data better reflects real clinical conditions. The merge also makes it possible to analyze and compare normal cells and leukemia-related cells within a single dataset instead of treating them as two independent problems.

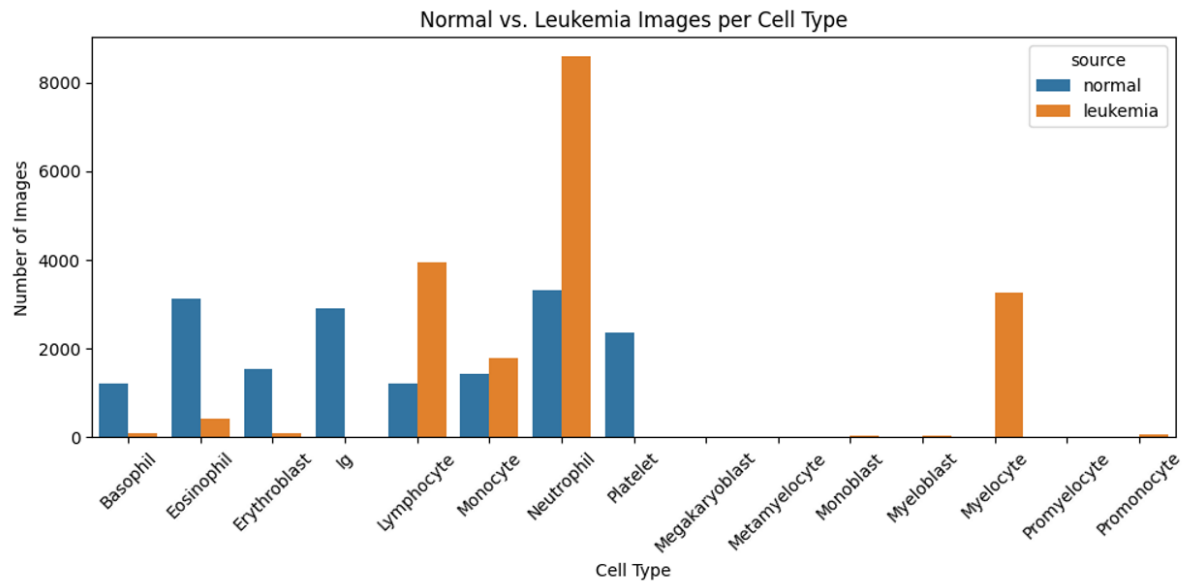
3.2 Advantages of merging the datasets

One important advantage is that the merged dataset contains more data and more variation. This helps machine learning models learn more general patterns and reduces the risk of overfitting to only one dataset. Another advantage is practical relevance. Since normal and leukemia cells coexist in real blood samples, a merged dataset is closer to real-world situations and therefore more meaningful for medical applications.

Merging the datasets also allows different types of classification tasks. For example, the data can be used to classify detailed cell types, or to perform simpler tasks such as distinguishing between mature and immature cells.

Finally, the merge makes it easier to compare similar cell types from normal and leukemia data, helping to better understand how disease affects cell appearance.





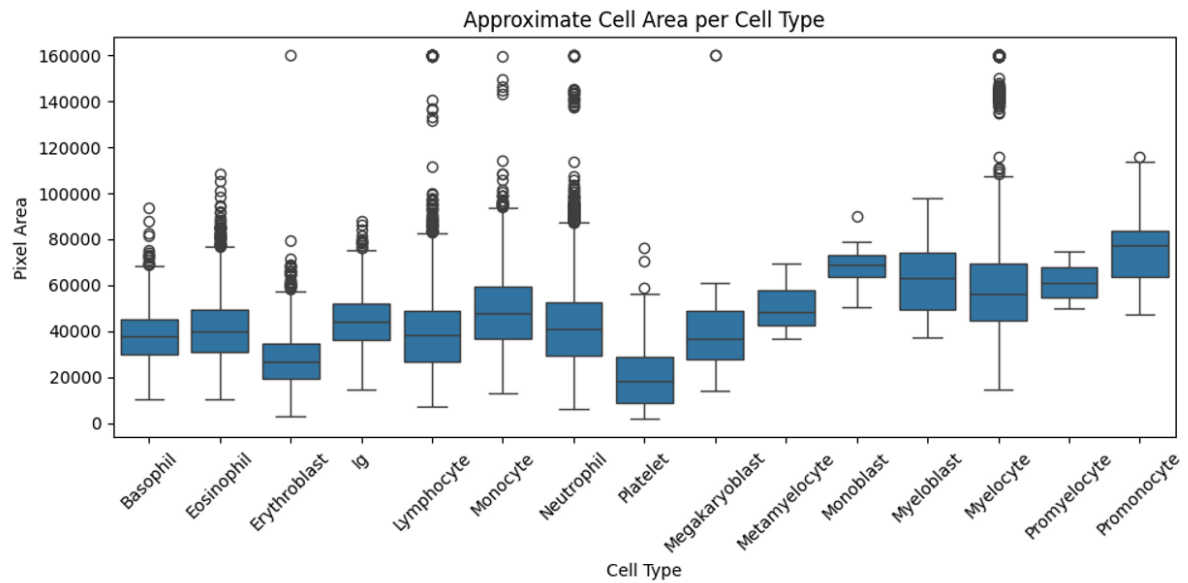
3.3 Disadvantages and challenges

A major challenge of merging the datasets is class imbalance. Some leukemia cell types appear much more frequently than others, which can bias the model if not handled carefully. Another issue is uncertain labels, especially for immature cells. These cells change gradually during maturation, so it is often difficult to clearly distinguish one stage from another. This means that the labels are not always perfectly accurate. In addition, the two datasets may differ in image quality or staining methods, which can introduce unwanted differences that are unrelated to the actual cell types.

3.4 Why merging is still useful

Even with these challenges, merging the datasets is still useful and commonly done in research. As long as the limitations are clearly explained and the results are interpreted carefully, the merged dataset provides valuable insights.

The goal of merging is not to simplify the data, but to better represent the complexity of real medical images.



3.5 Summary

In summary, merging normal and leukemia blood cell datasets:

- increases data diversity,
- better reflects real blood samples,
- supports different classification tasks,
- but requires careful handling of imbalanced and uncertain labels.

Overall, the merged dataset forms a solid basis for further analysis and model development.